

# Project Report

## *HiUni: Campus App for the University of Hildesheim*

Team:

| Name             | Student ID |
|------------------|------------|
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## 1. Use Case

- **Problem statement:** Students at the University of Hildesheim use a dozen digital services every day: canteen menu, LSF grade overview, Learnweb submissions, library room booking, university sports, campus cinema, university mail, timetable. They all live on separate websites, without a central overview and without a universal login. Anyone who checks food, submissions or mail several times a day signs in to each service anew, and does so on a smartphone that most of these sites were never made for.
- **Motivation:** We wanted the services we ourselves need several times a day in one place, and in a form that other students can use too. Comparable apps exist (e. g. Studo), but they do not cover the University of Hildesheim in the depth that really simplifies everyday life: no LSF grades, no group-room booking, no university mailbox. And the project was the chance to try out the MSE content (Android architecture, Compose, Lifecycle, persistence, testing) on a problem that affects us directly.
- **Real-world relevance:** The target picture from our project description reads: “A student at the University of Hildesheim should know within ten seconds of launching the app what she has to do today, what is on offer in the canteen, which mails are unread and which submissions are due, without having to sign in anywhere again.” Today this takes five websites and four logins.
- **Deliberate scope limits:** HiUni is *not* a replacement for LSF and Learnweb: the app aggregates public content and login sessions; where a real administrative action is required (sports-course registration), it points to the official page. It is also not a rating platform and not an application run by the university IT department, but a private student project within the MSE module, not coordinated with the university IT. Cross-platform was deliberately not chosen: native Android was the course requirement and the learning goal.

## 2. UX Design and Target Group

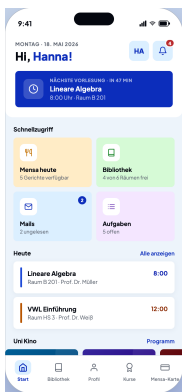
### 2.1. Target Group

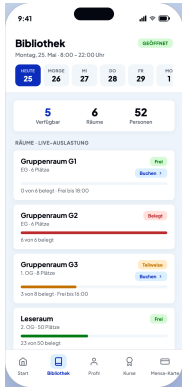
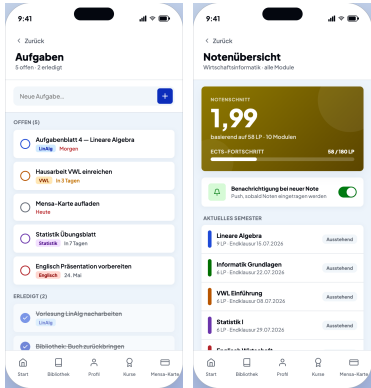
- **Primary users:** Students at the University of Hildesheim, across all semesters and degree programs. Deliberately scoped broadly, because canteen, mail, timetable and library are independent of the degree program.

- **Secondary users:** Staff with a teaching assignment who eat in the canteen or check their university mailbox; usable, but not the design focus, since exam, grade and submission features are tailored to students.
- **Explicitly not the target group:** Students at other universities (all endpoints, scrapers and credentials are specific to the University of Hildesheim) and iOS users.
- **User needs (today's workaround in parentheses):** Know what is on today (LSF timetable in the browser, no daily overview); see the canteen menu (separate STW-ON website); check unread mail (dedicated webmail login with the RZ account); keep an eye on submissions (Learnweb, CAS login on every visit); get an overview of grades and average (LSF grade report, aggregate in your head); find and book a free group room (ubwww with its own session handling); find a sports course (HSP program page, booking behind its own login).
- **Usage scenario (our own phrasing of the target picture from Section 1):** A student opens HiUni on the bus in the morning. On the home screen she sees the next lecture with a countdown and room, below it today's canteen offer, the unread mails and the next submission due. She taps the submission, sees the deadline, and books herself a group room for the afternoon on the way, without signing in to four portals.

## 2.2. Mockups

- The design reference was a high-fidelity HTML/CSS mock (*Uni Hi.html*) with 17 screens, produced with *Claude Design* and sharpened by hand; the AI reflection on it is in Section 4.2. From it we translated the color system (OKLCH values as sRGB constants), the typography scale, the corner radii and the home layout into the Compose theme.
- The three core workflows in the following table come from it; the images are rendered directly from the mock.

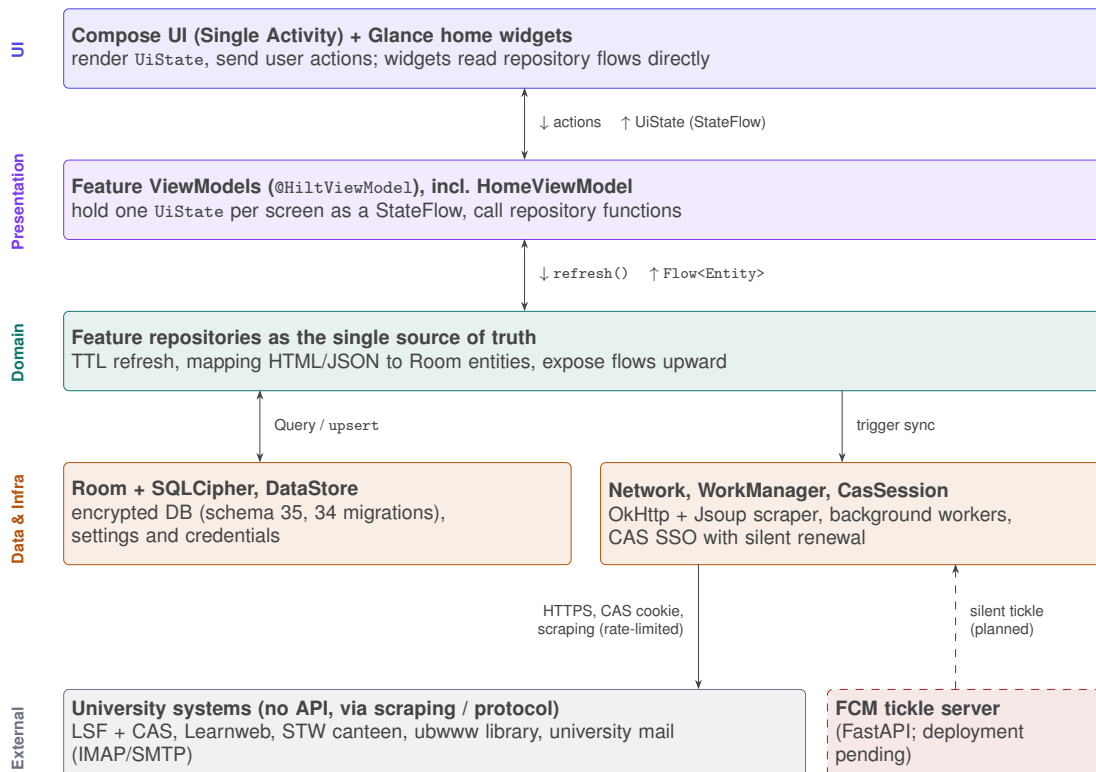
| Screen                | Mockup                                                                              | Addresses which user need?                                                                           |
|-----------------------|-------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------|
| Home (daily overview) |  | Daily overview without multiple logins in one screen: what is on, what there is to eat, what is open |

| Screen               | Mockup                                                                             | Addresses which user need?                                                                    |
|----------------------|------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------|
| Library / group room |   | Find a free study room and book it directly, instead of operating the ubwww page on the phone |
| Tasks and grades     |  | Submissions and grades at a glance, without logging in via CAS on every visit                 |

## 2.3. Design Rationale

- **Deliberately omitted or simplified:** The mock had 17 screens; we started with eight navigation destinations (Home, Calendar, Canteen, Cinema, Library, Mail, Settings, About). Exams, grades, sports and the push center followed later; study groups and a campus map are not built: the remaining effort did not fit the time budget, and study groups without a backend would have stayed a local note list. We dropped the in-app tweaks panel (accent color, greeting text) because it would have duplicated the settings. The About screen deliberately remained a minimal stub.
- **Design system instead of stock components:** We rebuilt pills, badges and the “pillow” indicator of the bottom navigation from Surface and Text by hand, rather than taking Material 3 components with a different look. It costs more code, but it keeps spacing, radii and color roles consistent across all screens.
- **A UX decision against the AI proposal:** For the day view of the calendar the AI generated a full hourly grid from 8:00 to 22:00 with per-event position calculation. In the pair review we rejected this as over-engineered for that project phase and reduced it to an agenda list; the rest of the feature wave mattered more than a pretty day view. In a later polish phase a color-coded hourly grid (8:00–18:00) did arrive after all. So the proposal only came too early, and in review we have since also checked whether something fits the current project phase.

### 3. System Architecture Diagram



How to read it: arrows pointing down are calls, arrows pointing up are data streams (flows).

#### 3.1. Architecture Justification

- Why this structure?** The app is strictly layered (Compose → ViewModel → Repository → Room/network), but packaged *feature-first*: each of the 21 feature packages bundles entities, DAOs, repositories, ViewModels and screens in one place. The repository is the single source of truth; the UI never reads from the network, only from Room flows, which is why every screen has content instantly and the app stays usable offline. Cross-feature access is forbidden by convention, with two documented exceptions: *feature.home* (combines eleven repositories) and *feature.widgets* (Glance has no ViewModels). Because five external systems are integrated via HTML scraping (LSF, Learnweb, ubwww, unifilm, university sports), the HTTP client, the politeness throttle, the CAS session and the background workers sit outside the features; otherwise every scraper would carry its own rate-limit and session logic. The FCM tickle server is deliberately a separate, content-free box: it knows only device tokens and sends a pure “go check” impulse, so that no credentials or mail content ever touch a foreign server. It is implemented and tested but not yet deployed to production; until then the mail sync runs via the periodic WorkManager workers. Security-critical data lives in a SQLCipher-encrypted Room database with a key from the Android Keystore.
- Rejected alternatives:** (1) *Layer-first with several Gradle modules*: a single module builds faster at our size and the feature ownership does not fray; the price is missing compile-time isolation, but the packages map one to one. (2) *Retrofit + Moshi*: not worth it for a handful of JSON endpoints alongside mostly HTML scraping; OkHttp plus kotlinx.serialization suffice. (3) *Koin instead of Hilt*: saves boilerplate, but costs the compile-time check of the dependency graph. (4) *One shared calendar table*: the discriminator approach breaks the feature separation, so pinned entries are now snapshots in their own table. (5) *AGP 9: Hilt*

broke completely with it, back to AGP 8.7 + Kotlin 2.0 + KSP1.

- **Walk-through (scenario from 2.1: “open the app in the morning”):** (1) The tickle server is meant to send a silent, high-priority FCM data message with no payload every 15 minutes (server built, deployment pending; currently the periodic worker starts instead). (2) The `HiUniMessagingService` receives it, obtains its dependencies via a `Hilt EntryPoint` (Firebase instantiates it itself), checks preconditions and enqueues an *expedited* `MailPushSyncWorker`; the service does not do the work, because its process can end immediately. (3) The worker calls `EmailRepository.refresh()` and classifies network errors as `retry` and auth errors as `failure`; in parallel the `LsfSyncWorker` obtains a service ticket via `CasSession` with silent renewal. (4) All HTTP calls go through the shared `OkHttp` client with a cookie jar and a `PolitenessInterceptor`; Jsoup parses HTML, `kotlinx.serialization` the canteen response. (5) Results are upserted into Room, a single write that all observers hang off; the DAOs emit new flows, the repositories pass them on. (6) The `HomeViewModel` combines the flows from eleven feature repositories (calendar, canteen, mail, tasks, ...) into a `HomeUiState`. (7) The `HomeScreen` collects it with `Lifecycle` and draws the banner, canteen card and mail badge; the timetable widget reads the same flow via the widget `EntryPoint`.

### 3.2. Third-Party Libraries

In one sentence: Hilt wires the app together, Room and SQLCipher store it encrypted, OkHttp and Jsoup fetch and parse the university pages, Jakarta Mail speaks IMAP/SMTP, `kotlinx.serialization` and `biweekly` read JSON and iCal, Coil loads images, Firebase provides the push transport, and MockK/Turbine/Robolectric carry the tests. Everything below solves exactly one task that you would otherwise have to build yourself.

*Alongside the platform building blocks from Google/AndroidX (Compose, Navigation, Lifecycle, DataStore, WorkManager, Glance, Biometric, Window/WindowSizeClass, Fragment for BiometricPrompt, SplashScreen, Palette for poster colors, Material Icons, Google Fonts) and the Kotlin coroutines, which we count as part of the architecture decision rather than as third-party libraries, the following third-party libraries are in use. In addition, but without effect on the domain logic: Timber (logging) and LeakCanary (debug build only).*

| Library                                   | Purpose                                                                                                                | Why this one (over alternatives)?                                                               |
|-------------------------------------------|------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------|
| Hilt / Dagger 2.56                        | DI, <code>@HiltViewModel</code> , <code>@HiltWorker</code> , <code>EntryPoints</code> for widgets and the push service | Compile-time check of the dependency graph instead of runtime surprises                         |
| Room 2.6.1 (+ KSP)                        | SQLite access, one <code>AppDatabase</code> , entities and DAOs in the feature, versioned migrations                   | Type-safe queries and Flow integration; KSP1, because Room 2.6.1 does not compile with KSP2     |
| SQLCipher (sqlcipher-android 4.6.1)       | Encryption of the entire Room database                                                                                 | Mail cache, grades and exam data sit on disk; prefs alone leave the DB open                     |
| OkHttp 4.12 (+ logging interceptor)       | HTTP client for scrapers and APIs, cookie jar, cache, politeness interceptor                                           | Cookies, pool and cache built in; with <code>HttpURLConnection</code> that would be manual work |
| Jsoup 1.18.1                              | HTML parsing for LSF, Learnweb, ubwww, unifilm, university sports                                                      | Robust CSS selectors on broken university HTML instead of regex                                 |
| Angus Mail 2.0.3 / Jakarta Mail API 2.1.3 | IMAP and SMTP client of the mail feature                                                                               | Protocol-correct including IDLE and SPECIAL-USE; a custom parser would be weeks of work         |

| Library                                        | Purpose                                                                            | Why this one (over alternatives)?                                                                       |
|------------------------------------------------|------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------|
| kotlinx.serialization 1.7.3                    | Type-safe DTOs for JSON endpoints (canteen, TMDB)                                  | Kotlin-native, no reflection overhead; Retrofit/Moshi only pay off with more endpoints                  |
| Coil 2.7.0 (coil-compose)                      | Asynchronous loading of the cinema posters                                         | Cache, concurrency and Compose integration ready-made                                                   |
| Firebase Cloud Messaging (firebase-bom 33.7.0) | Reception of the silent tickle data messages from our push server                  | The only way out of Doze without polling; only the transport is used, no analytics                      |
| biweekly 0.6.8                                 | iCalendar parsing (RFC 5545) of the Learnweb feed as a second source for deadlines | Correct RRULE and EXDATE expansion is a project of its own                                              |
| MockK 1.13.13, Turbine 1.1.0, Robolectric 4.13 | Unit tests: Kotlin mocking, flow assertions, Android tests on the JVM              | MockK understands Kotlin idioms, Turbine makes StateFlow tests readable, Robolectric saves the emulator |

### 3.3. The Tickle Server: Concept and Status

Noticing new university mail quickly collides with Android: from Doze mode an app may not check by itself every minute. The usual way out would be a server that watches the mailboxes and, on new mail, sends a push message with content, but then credentials and mail content would sit on foreign infrastructure. Our *tickle* server avoids exactly that: it knows only FCM device tokens, that is, only “which device can I push to”, and sends a silent, content-free, high-priority data message every 15 minutes (`{type: sync_tickle}`). That wakes the device from Doze; the app realizes “time to check” and fetches the mails itself via IMAP, with credentials that live only on the device. The model is deliberately built like WhatsApp’s: the push infrastructure knows only that something is probably pending, not what. If the server fails or is compromised, at most device tokens are affected, no mailboxes. On the server side there are token purge (invalid tokens are dropped automatically), a rate limit and an X-Api-Key. The server (FastAPI, around 300 lines, Docker-ready) is implemented and tested but not yet deployed to production; until then the periodic WorkManager worker handles the mail refresh. The FCM keys (the server’s service account, the app’s `google-services.json`) are deliberately not in the repo but excluded via `.gitignore` and injected through environment variables or CI secrets; the repo holds only an `.env.example` as a template. So a checkout intentionally contains no push credentials.

## 4. Developer Diary

This section shows the course of development: first the milestones, then the collaboration with the AI with its successes and failures.

### 4.1. Development Milestones



| #  | Date       | Activity                                                                                                                | Result                                                                                            | Issues                                                                        |
|----|------------|-------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------|
| 1  | 2026-05-23 | Foundation: Gradle, Compose, Hilt, Room, DataStore, WorkManager, AdaptiveScaffold, 7 ADRs, stubs                        | assembleDebug and lintDebug green on the same day                                                 | KSP2 broke the Room annotations, rollback to KSP1                             |
| 2  | 2026-05-24 | Calendar CRUD (list, day, week, reminder); canteen via the STW-ON API                                                   | Both done, test pattern for repositories in place                                                 | AI DTO of the canteen API with invented field names; day view over-engineered |
| 3  | 2026-05-26 | Library group rooms: floor plan, slot picker (max. 2 h), session, book and cancel                                       | Booking flow in the app instead of the browser                                                    | Session-fixation bug in the real ubwww; report to the library pending         |
| 4  | 2026-05-31 | Scraper wave: cinema, library occupancy, mail (IMAP/SMTP); sports followed at the end of June                           | All three in; library/sports (Johann), mail/cinema (Kjell)                                        | IMAP folder names differ per server; fix via SPECIAL-USE                      |
| 5  | 2026-06-27 | Tablet foundations (rail, drawer, multi-pane); on the side, as an experiment, P2P canteen ratings (Ed25519, Ktor relay) | Adaptive layouts everywhere; the ratings stayed an unmerged experiment and were dropped           | BottomSheet on BottomSheet did not stack; percentage value truncated          |
| 6  | 2026-06-29 | Learnweb in four stages: CAS SSO, course list, submission deadlines in the calendar, iCal                               | Complete in one day, reminders 3 d / 1 d / 2 h                                                    | Reminders first in the generic event category, own one from 17.07.            |
| 7  | 2026-07-13 | Five Glance widgets plus a design kit and deep links, Ultracode in parallel in worktrees                                | All five shipped, around 600 lines of layout replaced by the kit                                  | Glance limit of 10 children undetected for four days (silent exception)       |
| 8  | 2026-07-16 | Grades and average from the LSF grade report, offline marking                                                           | Grades screen live, test suite 140 → 289                                                          | Fixture with real exam data, now synthetic with a data-protection assertion   |
| 9  | 2026-07-16 | FCM tickle server (FastAPI) for mail push without credentials or content                                                | Implemented, Coolify deployment prepared (service account as an env variable); deployment pending | Rate limit and token age added only afterwards                                |
| 10 | 2026-07-17 | Grade report linked to courses (course number), semester anchor for the icon, Glance fix, Learnweb channel              | Four commits in one day, 475 tests, 0 failures                                                    | Precedence of manual grade over grade report secured                          |

## 4.2. AI Interaction Log: Critical Incidents

The whole project was created with *Claude Code* (Anthropic, model Opus 4.x with a 1M-token context), but deliberately not in a question-and-answer style. The division of roles has been in an

AI\_USAGE.md in the repo since day one: the AI wrote around 80 % of the raw code, we made the architecture, library and scope decisions. The seven ADRs came about the same way: phrasing drafts from the AI, decided in conversation. Two operating modes of Claude Code were decisive here and go beyond mere dictation: the *Superpowers* skills in planning and the *Ultracode* mode in implementation.

**Planning ran AI-assisted too.** The *Superpowers* skills are reusable work instructions that impose discipline on the agent before it writes code. The ratings feature makes this traceable: from a brainstorming dialogue a 654-line design spec emerged on 28.06., iterated over the afternoon in five commits, with a section “Resolved points (decided in brainstorming)” and the removal of an entire phase (LAN sync via mDNS) after research showed that Eduroam blocks multicast. From it the *writing-plans* skill produced an implementation plan of 3 344 lines: 7 phases, 34 tasks, each a TDD three-step (failing test, implementation, commit). What matters to us is the separation: the spec is the document we decided on, the plan only its mechanical unfolding.

**Implementation ran twice in *Ultracode* mode**, that is, as multi-agent orchestration across separate Git worktrees. The first use accompanied the later-dropped ratings experiment (28.06.): two agents built in parallel, plan-driven, and after each phase a review ran across the whole branch. Fully traceable in the history of `main` is the second use, the widgets (13.07.): between the Glance foundation and the last merge lie, according to Git timestamps, 63 minutes with 19 commits and around 2 900 inserted lines. The sequence is readable in the history: foundation sequential, three of the five widgets in their own agent worktrees (the main session built two directly), then a shared design kit and three further worktrees that switched the finished widgets over to it; their merges have *negative* net diffs, because the kit replaces duplicated layout. A human merged throughout: all merge commits carry the same author, a manifest conflict is documented there and was resolved by hand. On top of that came an automated `/code-review` per branch before the merge. The following five incidents are the ones we learned the most from.

| # | Milestone   | What happened                                                                        | AI's output (brief)                                                                                                                                                        | Your action / validation                                                                                                                          |
|---|-------------|--------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------|
| 1 | 3 (Library) | Two Set-Cookie headers in the booking flow; session logic analyzed with the AI       | ubwww rotates the PHPSESSID after the CAS login but does not invalidate the old session: “one booking per day” can be bypassed, leaving orphaned reservations              | Reproduced with curl and DevTools in a bug document; the client now takes the <i>last</i> cookie. Report to the library IT pending                |
| 2 | 7 (Widgets) | In the exam-countdown widget the room was missing in the large size, without a crash | MetaChip placed icon, spacing and text as three siblings in the row: 11 children where Glance allows 10; every render threw a silent <code>IllegalArgumentException</code> | Chip wrapped into a row wrapper (one child), <code>WidgetLimits</code> with the limit and a unit test; the AI does not know such framework limits |
| 3 | 5 (Ratings) | Approval percentage of the aggregated canteen ratings                                | $(\text{positive} * 100) / \text{total}$ as the “correct” formula, i.e. integer division that truncates instead of rounding                                                | Only caught while writing the tests; fix and edge-case tests in the ratings experiment, test-first since then                                     |
| 4 | 2 (Canteen) | DTO for the STW-ON canteen API, for which there is no public documentation           | Invented field names (prices, notes instead of price and a nested tags): compiled cleanly, returned empty lists                                                            | Live call with curl, JSON sample into the repo, DTO corrected. Since then, inspect first, then generate                                           |



| # | Milestone   | What happened                                                            | AI's output (brief)                                                                            | Your action / validation                                                                                               |
|---|-------------|--------------------------------------------------------------------------|------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------|
| 5 | 7 (Widgets) | Five widgets in one day, Ultracode in parallel in separate Git worktrees | Each worktree delivered a working widget, but with its own style: colors, paddings, structures | Throughput impressive, then a unification pass to one widget theme. A design kit belongs <i>before</i> the parallelism |

### 4.3. Project Documentation

The development process is documented in the repo, not only in the code. The most important artifacts:

- docs/adr/ **(ADR 0001–0007)**: one architecture decision each on feature-first packages, single-activity Compose, Hilt, one AppDatabase, library strategy, “no unified calendar” and AGP 8.7; the AI supplied drafts, the team decided.
- docs/process/ **(01–07)**: project description, architecture overview, chronological engineering log, AI workflow (disclosure plus prompt log and reflection), build instructions, team/contributions, interim pitch.
- docs/superpowers/ **spec and plan**: the 654-line design spec and the 3 344-line implementation plan derived from it for the ratings feature (see Section 4.2).
- **Repo root**: AI\_USAGE.md (living AI disclosure, updated per commit wave), CHANGELOG.md, docs/FEATURES.md (feature status), docs/DEVELOPMENT.md (internal cookbook) as well as the concept and plan files of the initial phase (HIUNI\_KONZEPTE, -LIBRARIES, -REFACTOR-PLAN).

## 5. Android Features Report

### 5.1. Features Used

The app uses the following platform building blocks. “Why” names the concrete requirement in each case, not just the technology.

| Feature                                    | Where it's used (screen/class)                                                                                | Why this feature was needed                                                                                                                             |
|--------------------------------------------|---------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------|
| Activity lifecycle, intents and deep links | MainActivity (FragmentActivity, singleTask), onCreate/onNewIntent, Widget- and NotificationDeepLinkController | Single activity with one Compose NavGraph; widget, notification and NFC intents land in the running instance and are mapped to a navigation destination |
| Local database: Room with SQLCipher        | AppDatabase (schema 35), entities and DAOs per feature, Migrations.kt                                         | Offline-first: every screen reads from the DB, never from the network. Encrypted, because the mail cache and grades sit in it                           |
| Schema migrations                          | 34 additive Migration objects, schema export to app/schemas                                                   | No destructive migration: an app update must not delete user data                                                                                       |
| DataStore (Preferences)                    | SettingsDataStore                                                                                             | Settings, sync intervals, navigation order and refresh timestamps, asynchronous without SharedPreferences blocking                                      |

| Feature                              | Where it's used (screen/class)                                                                         | Why this feature was needed                                                                                         |
|--------------------------------------|--------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------|
| Encrypted preferences + Keystore     | CredentialsManager, DatabaseKeyProvider (Encrypted SharedPreferences)                                  | Mail credentials and the 32-byte DB key must not sit in plaintext; master key in the Keystore                       |
| Biometrics                           | MailLockGate (BiometricPrompt, BiometricManager)                                                       | The mail tab shows other people's communication, unlocked by fingerprint or device code; hence the FragmentActivity |
| Runtime permissions                  | OnboardingScreen, RemindersSettingsScreen (POST_NOTIFICATIONS), canScheduleExactAlarms(), NFC optional | Notifications from API 33, exact alarms opt-in from API 31; the app must run without either permission              |
| Notification channels and categories | NotificationPresenter, NotificationCategory plus an in-app push center                                 | Events, grades, mail and system separately mutable; the in-app center logs independently of the OS permission       |
| AlarmManager                         | NotificationScheduler (setExactAndAllowWhileIdle)                                                      | Event, exam and submission reminders must fire on the right minute even in Doze                                     |
| WorkManager                          | LsfSyncWorker, SportSyncWorker, MailPushSyncWorker, PushRegistrationWorker (@HiltWorker)               | Guaranteed background work despite Doze and battery optimization, with a network constraint and backoff             |
| Firebase Cloud Messaging             | HiUniMessagingService, MailTickleHandler                                                               | Without push the app would have to poll every minute; the silent data message wakes it but carries no content       |
| App widgets with Glance              | feature.widgets (five widgets), WidgetHiltEntryPoint                                                   | Timetable, canteen, tasks and exam countdown on the home screen without launching the app                           |
| NFC (IsoDep)                         | NfcScanController, MensaCardReader                                                                     | There is no API for the student card's balance, so ISO-14443-4 APDUs directly from the card                         |
| WindowSizeClass / adaptive layouts   | AdaptiveScaffold, NavigationType                                                                       | The same program on phone and tablet: bottom bar, rail or drawer by width, including multi-pane                     |

## 5.2. Implementation Details

- NFC reader for the student card.** A singleton (NfcScanController) mediates between ViewModel and Activity via two SharedFlows (detected tags, navigation request) and a StateFlow scanning. The MainActivity observes scanning in the lifecycleScope and turns on the foreground dispatch only then; onPause always disables it, onResume reactivates it only during a running scan, so the app does not “steal” a card that comes near the device outside the payment flow. Registration is explicit with techLists = [IsoDep], otherwise another tag technology blocks IsoDep (“only one TagTechnology can be connected at a time”). Two entry points: an external ACTION\_TECH\_DISCOVERED intent on a cold start (the controller sets scanning itself and navigates to the card screen) and the scan button in the screen. At the protocol level: DESfire over ISO-7816-4-wrapped APDUs, multiple frames via status 0xAF, UID from the anticollision rather than via a vendor command (current cards answer that with “illegal command code”); IsoDep.connect() up to three times with 80/160

ms backoff.

- **Five Glance widgets.** Glance does not render into our process but into RemoteViews, so almost no app architecture applies here: a GlanceAppWidget cannot be an `@AndroidEntryPoint`, so the widgets obtain their four repositories via an `@EntryPoint` and `EntryPointAccessors.fromApplication`; without a `ViewModelStore` they collect the Room flows directly. Updates run not via the AppWidget timer (`updatePeriodMillis=0`) but via the flow emissions; interactive buttons run via `ActionCallback` with a subsequent `update()`. Colors must come as `ColorProvider(day, night)`, otherwise dark mode breaks, and a custom widget theme replaces `MaterialTheme`. The limit of 10 direct children per container has been a named constant with a test in the code since the silent rendering bug.
- **Push “tickle” plus background sync with CAS SSO.** The server sends no payload, only a type marker (server done, deployment pending; the app-side receive path is fully implemented). The decision logic (“should this tickle trigger a sync?”) is decoupled from Firebase and runs as a JVM test. The `FirebaseMessagingService` is instantiated by the SDK, not by Hilt, hence its own `EntryPoint`; it reads preconditions blocking, because the callback runs on a background thread anyway. Work happens in an *expedited* `MailPushSyncWorker` (`RUN_AS_NON_EXPEDITED` as a fallback, `KEEP` against tickle bursts), because the service process ends after the callback; the worker re-checks the credentials and classifies errors into `retry` and `failure`. The periodic `LsfSyncWorker` works through courses, timetable, exams and the grade report with a 400 ms throttle; auth errors are retried up to twice so the CAS session renews silently, a parse error does *not* abort the run but produces a deduplicated warning in the push center (“exam dates outdated”). The most critical detail sits in the session: CAS binds the ticket-granting cookie to the user agent of the issuing `WebView`; without replaying it the server deletes it on the next request.
- **Edge cases.** *Notification permission denied*: the entry always goes into the in-app push center, the system notification only with granted `POST_NOTIFICATIONS`. *No exact alarms*: fall back to `set()`. *Offline*: the `ConnectivityObserver` requires *internet and validated* (no captive-portal false alarm) and debounces 1,5 s, a banner above the `NavHost` shows the state centrally, a label marks stale data (offline or refresh older than 6 h). *No NFC chip*: in the manifest `required=false`, the adapter may be `null`. *Configuration change*: deep-link intents are voided after being consumed (`intent.action = null`), the scaffold reacts live to the new `WindowSizeClass`. *Doze*: exact alarms for reminders, expedited work for push syncs.

### 5.3. AI’s Role in Android-Specific Code

- **Where the AI was reliable:** on well-documented AndroidX standard (Hilt modules, `@HiltWorker` with `@AssistedInject`, Room DAOs and migration scaffolds, Compose screens with `collectAsStateWithLifecycle`, `WorkManager` constraints) it mostly produced correct and up-to-date code on the first try. Outdated patterns (`AsyncTask`, `LiveData`, fragment transactions) we practically never saw in three months.
- **Where it regularly went wrong:** on poorly documented hard limits. The classic case is the Glance limit: plausible code, the widget renders, only one element is missing, the error lands silently in the log. Second, the binding of the CAS cookie to the `WebView` user agent, a detail from no documentation, findable only via log analysis. Third, real NFC behavior (reconnect retry needed, a vendor command aborts the session), resolvable only on the device.
- **Hallucinations:** no invented Android APIs, but invented *foreign* schemas; the guessed field names of the canteen API are the same error class: syntactically correct, semantically wrong, silently ineffective.
- **Consequence for the workflow:** before generating code against an unknown interface, there is a live call with a saved sample, a framework limit once hit is captured as a named constant with a test, and everything around hardware or system services (NFC, biometrics,

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Doze, notification permissions) we check manually on two fixed devices (Pixel 7a and Galaxy Tab S6), because neither a unit test nor the AI can say anything about it.

## 6. Lessons Learned

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- **Technical:** scraping is a permanent risk, not a one-off problem, so changed HTML structure is its own error class that does not abort the sync but warns in the push center. We found security details by looking closely, not by planning: cookie selection in the library client, the user agent on the CAS ticket, an old plaintext fallback for the database key, since removed. 34 additive migrations are maintenance effort, but in three months no lost user data.
- **Architectural:** feature-first with two documented cross-feature exceptions held up; a tempting third one was every time a sign of a missing core service. Offline-first via TTL refresh plus a staggered warmup was the single best decision for the perceived quality: no screen shows a skeleton on switching. Also, boring-and-stable tools over the newest versions: the retreat from AGP 9 to 8.7 and from KSP2 to KSP1 cost one day and probably saved weeks.
- **UX:** a high-fidelity mock saves discussions but tempts you into scope: 17 screens in the mock, 8 at the start. Function before looks was right in the build-up phase (agenda list instead of an hourly grid), the visual debt could be paid down later. Small but effective: we *removed* the “add to calendar” feature for canteen menus after rolling it out, because pinned dishes showed up in the home banner as the “next lecture”.
- **AI-related:** “architecture and scope by hand, raw code generated, review mandatory” worked. Claude Code was valuable on uniform code (widgets, DAOs, migrations, test scaffolds) and in the analysis of foreign systems; the session-fixation bug we would probably never have found by hand. It was costly on silent errors: wrong calculations and invented schemas compile cleanly. Exactly against that the Superpowers skills paid off, because TDD is enforced as the default. Ultracode mode raises throughput enormously but produces style divergence; a design kit must come *before* the parallelization.
- **Again / differently:** again: ADRs before the first code, tests as the standard rather than an extra, live inspection of every interface, a single Gradle module. Differently: notification categories fine-grained from the start (the Learnweb reminder ran for a good two and a half weeks under the wrong one), UI texts as string resources from the start instead of German literals, and platform limits like the Glance limit pinned down as a test immediately.

## 7. Future Work

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- **Possible refactoring:** move UI texts to `strings.xml`; the German literals in the composables block multilingualism and text tests. Build out the About screen (a stub of around 30 lines without licenses, build info and an AI-usage note) and put the planned trigger for theme easter eggs there. Remove the read-only compatibility path for the historical plaintext database key once no old installations exist any more. Systematize the Glance limit check: the test covers only the countdown widget, one per widget and size would make sense. And complete the deep links of the global search: today a hit opens only the tab, not the entry.
- **Possible extensions:** first deploy the finished FCM tickle server to production (the Coolify configuration is ready), so that the mail-push path runs not only in tests. Then study groups

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with join/leave, filters and calendar mirroring (no code exists for it). A campus map as a vector composition with a location pin and filters by lecture hall, library, canteen and sports. A digital student card with a QR code and semester-ticket info. Favorites for library rooms with a notification as soon as a favorite room becomes free. Reminders for due tasks (the calendar pattern transfers, it is just not wired up). Local canteen ratings as a personal running average. And the open wiring in the cinema feature: the trailer URL is in the database, the call is missing.